

HARSH SHARMA

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SUMMARY

Backend & Infrastructure Engineer with 3 years of experience building scalable systems, APIs, and developer tools across AI, EDA, Robotics, and Web3. Deep expertise in Python, Go, TypeScript, and distributed backend architectures with hands-on experience in Docker, Kubernetes, cloud infrastructure, and agentic AI integration. Proven record of end-to-end technical ownership, cross-functional leadership, and founding-team collaboration at multiple startups. Passionate about building simulation environments and infrastructure for AI agent training.

EXPERIENCE

Aurora System (csfirst) — Backend & Infrastructure Engineer | India Lead

Hybrid (Bangalore) | Nov 2024 – Present | Direct report to Founder; sole India-based engineer (team across USA & China)

- Agentic AI Extraction Pipeline: Built a multi-stage agentic pipeline combining web-scraping, LLM-driven analysis, and vision models to autonomously extract pin maps, power rails, and metadata from 1000s of semiconductor PDFs (Qualcomm project). Orchestrated agent workflows with retry logic, validation, and human-in-the-loop fallback.
- Parser Performance Optimization: Re-architected core AuroraDB parser with hash-map indexing and lazy evaluation, achieving 98% latency reduction (2-hour jobs completed in under 1 minute). Profiled and eliminated critical bottlenecks in the translation pipeline.
- EDA Format Translation Engine: Designed and built robust translators converting AuroraDB into industry-standard formats (Ansys AEDB, Cadence SPD, AAF, IPC-2581), enabling seamless interoperability across PowerSI, Sigrity, and Ansys EDT. Architected shared translation framework serving 5+ output formats from a single source.
- AuroraDB & EnhancedAuroraDB: Designed the structured database format for EDA workflows with full CRUD APIs. Extended into EnhancedAuroraDB supporting customer-specific re-routing and design modifications. Built GenAI-powered synthetic test case generator for large-scale regression validation.

Orangewood Labs — Team Lead & System Software Engineer (R&D), later Technical Advisor

Remote | Sep 2024 – May 2025 | Led engineering efforts directly with co-founders; represented company in customer & partner meetings

- Technical Leadership: Sole System Software Engineer and Team Lead for robot control platform. Elevated to Technical Advisor, attending customer demos and partnership meetings with leading robotics companies to close deals and share technology.
- Agentic Automation & Control: Developed AI-driven control algorithms (moveCircle, moveProcess, moveTrajectory) with autonomous error recovery and adaptive path planning. Implemented point cloud to robotic pose mapping with sensor-to-robot frame transformations for motion planning.
- Performance Optimization: Optimized Python and C++ control modules, cutting robotic arm runtime latency by 30%. Built automation ecosystem with custom debug tooling for Robot Operating System stability.

Cadence Design Systems — Software Development Engineer (R&D)

On-site (Noida) | Aug 2023 – Aug 2024 | EDA tooling R&D — Static Timing Analysis (STA), PBA/GBA, Remote Workspaces, CI/CD

- Built Remote Tempus Session (RTS) feature enabling decentralized, shared workspaces for large-loaded EDA designs, boosting team productivity and remote debugging. Integrated CI/CD automation with IBM Cloud for compilation and test orchestration.
- Engineered memory-swap optimization for design reloads. Made tool compatible with containerized Apple (NY) Docker systems using dual PID/UUID file management. Delivered customer-requested enhancements in PBA and GBA (Static Timing Analysis).

Ansys Software Pvt. Ltd. — R&D Intern

Hybrid (Noida) | Jan 2023 – Jul 2023 | Design Elaboration, Power Optimization, Parser Modernization

- Replaced legacy Interia parser with Verific Parser across 20,000+ lines, reducing build time by 10%. Built debugging tooling with GDB and productivity scripts for faster regression testing and analysis.

GoGlocal — Team Lead, Full-Stack Development (Contract)

Mumbai | Apr 2022 – Jul 2022 | Led development team; worked directly with founding members

- Led development of global e-commerce platform (React, Django, REST, AWS EC2/RDS) with multi-country marketplace integration (Amazon, eBay), international payments, and B2B expansion features.

TECHNICAL SKILLS

Languages & Systems: Python, Go, TypeScript, C/C++, Bash, Java, CUDA, Kotlin

Backend & APIs: FastAPI, Django, gRPC, REST, Node.js, WebSocket, Playwright

Infra & Cloud: Docker, Kubernetes, AWS (EC2/S3/RDS), Azure, Jenkins, CI/CD, Linux

Databases: PostgreSQL, MongoDB, Redis, MySQL, Snowflake

AI / Agentic / GenAI: Agentic Pipelines, LangChain, LLM Fine-tuning, RAG, Prompt Engineering, OpenAI APIs, Function Calling, Tool Use, Human-in-the-Loop, Multi-Agent Orchestration, Vision Models, Synthetic Data Generation, GenAI Integration, PyTorch, TensorFlow, Hugging Face, Vector DBs, NLP, Transformers

Tools & Platforms: Git, GDB, ROS, JIRA, Vim, IBM Cloud, Bitbucket

EDUCATION

Netaji Subhas University of Technology (NSUT) — B.Tech, Electrical Engineering | Delhi, India | GPA: 8.4/10 | May 2023 | Neural Networks & Fuzzy Logic, DSA, Big Data, Computer Networks, Data Warehouse Management

LEADERSHIP & RECOGNITION

- Founding-team collaborator at 3 startups (GoGlocal, Orangewood Labs, Aurora System) — direct access to co-founders, strategic input on product direction
- IEEE Outstanding Volunteer Award (Delhi Section, 2022) — organized hackathons, workshops, mentored juniors at NSUT IEEE Chapter
- Languages: English (Professional), Hindi (Native), German (Intermediate), Chinese (Basic)